Manual

Car Licence Plate Identification System

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Submission date:	January 24, 2026

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1 Getting Started

Welcome to the Smart License Plate Detection System

Dear User,

Welcome to your **Smart License Plate Detection System** developed at **Hochschule Emden–Leer**. This device recognises and logs vehicle license plates in real time.

The system starts automatically when powered on — no setup or configuration is required. Within seconds, it begins detecting and displaying vehicles on screen with accurate entry, exit, and duration details.

Key Highlights:

- Works entirely offline — no internet required.
- Designed for 24/7 airport gate operation.
- Automatically handles billing for stays beyond 15 minutes.
- Recognises exempt vehicles (police, ambulance, government).
- Compact, efficient, and easy to maintain.

"Technology that simplifies — accuracy that empowers."

Developed by **Gautam Ramesh**, **Siddhanth Sharma**, and **Shubhankar Dumka**

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1.1 List of Parts

Each component is plug-and-play, making the main unit simple to assemble, transport, and maintain. The setup is compact, energy-efficient (under 10 W), and fully operational without internet access — ideal for continuous airport use.

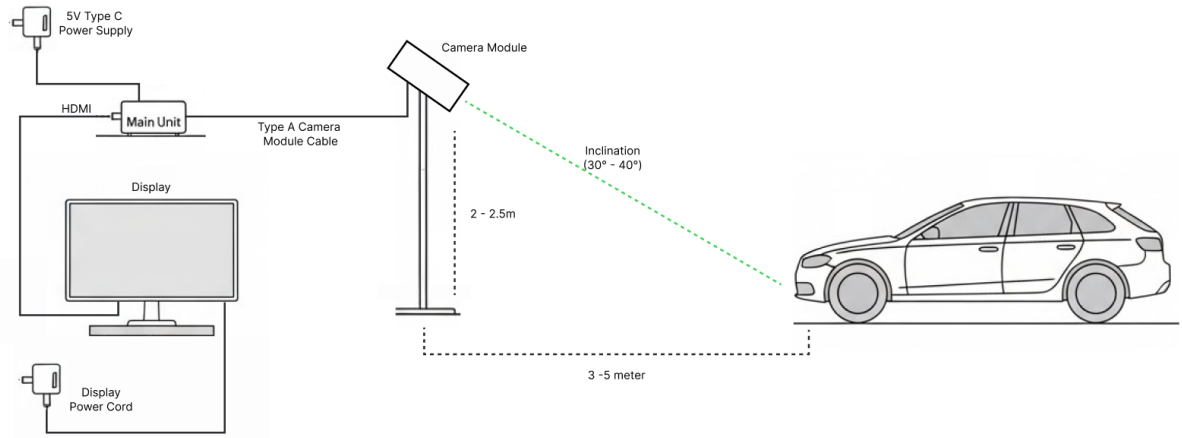
Table 1.1: Main Components of the System

S.No	**Part Name**	**Description / Function**	**Qty.**
1	Main Unit	Main central processing unit to enable detection, timing, and billing functions	1
2	Camera	Captures live vehicle images at the airport gate.Camera comes prebuilt with USB Type A cable	1
3	Camera Stand	Adjustable metallic stand for stable positioning.	1
4	Display	LCD monitor showing real-time license plate detections.	1
5	Power Cord 230V, 0.1A, 20W	Cable to connect display with power supply.	1
6	microSD Card	64 GB microSD card with OS preinstalled	1
7	Power Supply USB Type-C	5V/4A USB-C power supply adapter with USB-C Type outlet to power main unit.	1
8	HDMI Cable	Connects the display with main unit.	1
9	Mouse	Standard mouse with USB type A cable used for interfacing with the graphical user interface of the product	1

Figure 1.1: License Plate Detection Setup

2 Setting Up The Product

2.1 Product Setup



This section explains, in simple steps, how to prepare and start the **License Plate Detection System**. Follow the setup steps below to get the product ready for usage:

Step 1: Place and Connect the Hardware

- Place the **main unit** on a stable, well-ventilated surface.
- Mount the **camera** on the camera stand and point it toward the vehicle lane or entry gate.
- Connect the camera to the main unit using the USB Type A cable pre-built with the camera.
- Connect the **HDMI cable** from the main unit to the monitor.
- Insert the **microSD card** into the main unit.
- Connect the **microSD card** into the main unit.
- Connect the **mouse** to the main unit through its USB type A port.

Step 2: Start the Main Unit

- Plug in the **power cord** for the display and turn the display on using its power on button.
- Plug in the **power adapter** and connect it with the main unit using it's C-type DC-IN outlet.**Important:** Connect this power supply strictly with the DC-IN port at the left position on the back side, not the USB-C port at the right position on the back side.
- Wait for a few seconds for the main unit to start.
- A green LED indicator will turn on, indicating successful boot. If it isn't glowing, it indicates a problem with the main unit.

Step 3: Verify Operation

- After detection, any detected car license plates are logged with timestamps and saved. Additionally, the display will show the recognized license plate with entry time and subsequently, exit time at the time of exit.
- Adjust the camera height or tilt if the plates are not clearly visible.
- Ensure good lighting at the vehicle gate area. For accurate plate reading, maintain an illumination level between **250 lux (minimum)** and **900 lux (maximum)** — with an optimal range of **400–700 lux**. Avoid shining lights directly into the camera to prevent glare or overexposure.

Step 4: During and After Use

- The product runs continuously and does not need supervision.
- After use, if the product is to be shut down, the shut down button in the display interface can be clicked using the mouse to safely shut the product off.
- All data is stored safely for billing and reporting.

2.2 Camera Placement and Setup:

- Mount the camera about **2.0–2.5 metres above ground level**, facing the vehicle lane at a **30–40° angle**.
- Keep a distance of about **3–5 metres** between the camera and the point where vehicles stop or slow down.

- Avoid placing the camera directly opposite bright lights or reflective surfaces.
- Ensure there is enough ambient light. During night hours, add a small **LED lamp or floodlight** near the gate for better clarity.
- Keep the camera lens clean and free of dust, water spots, or scratches.
- Do not position the camera behind glass or near strong sunlight to prevent reflections.

Important Tips

- Place the device on a stable, flat surface with good ventilation.
- Make sure the **power supply (5 V / 4 A)** is connected firmly and the cable is not loose.
- Keep the **camera lens clean** for accurate plate reading.
- Do not block the openings on the aluminum case; it keeps the main unit cool.
- Avoid exposing the device to rain, moisture, or direct sunlight.
- Connect the HDMI cable to your monitor before switching on the power.
- Wait until the indicator light turns on — this means the entire sytem is ready.
- If the display stays blank, check power and camera connections first.

These simple steps help ensure your product starts safely and performs reliably every time.

Important to Know Before Use

- This main unit records number plates for time tracking and billing at the airport gate.
- Do not move or touch the camera once it is aligned. Misalignment can cause unreadable plates.
- If a plate is too dirty or not readable, you may need to type it in manually later.
- Emergency and official vehicles (police, ambulance, etc.) are recognised as **no charge**.
- Turn off the main unit only after normal shutdown. Unplugging power while it is running can cause data loss.

3 Main Function

3.1 Main Function

The **Airport License Plate Detection System** is a smart camera-based solution that automatically recognises and records the license plates of vehicles entering and leaving the airport area. It is specially designed for **German license plates** and helps manage short-term parking at the airport gate efficiently and fairly.

When a vehicle enters, the camera captures the license plate and stores it together with the date and time. When the same vehicle exits, the main unit detects it again and automatically calculates how long it stayed inside the airport area. If the duration is more than **15 minutes**, the main unit marks it for billing. If the stay is **15 minutes or less**, the parking is free of charge

Certain vehicles such as **emergency services (ambulance, police, fire brigade)** and **official government vehicles** are automatically identified and exempted from all charges. These special plates are pre-registered in the main unit's database, ensuring that authorised vehicles always pass freely without manual intervention.

All vehicle data, including plate number, entry time, exit time, duration, and payment status, are securely stored inside the device. This ensures transparency, traceability, and smooth traffic flow at both the entry and exit points of the airport

Figure 3.1: The Operator Dashboard showing the live video feed (left) and the real-time entry/exit log (right).

Key Highlights:

- Recognises and records German license plates automatically.
- Calculates parking duration and applies the 15-minute free rule.
- Automatically exempts emergency and government vehicles from charges.
- Works fully offline — no internet or network setup required.
- Stores all records locally for reporting, auditing, or payment.

3.2 Billing Mechanism

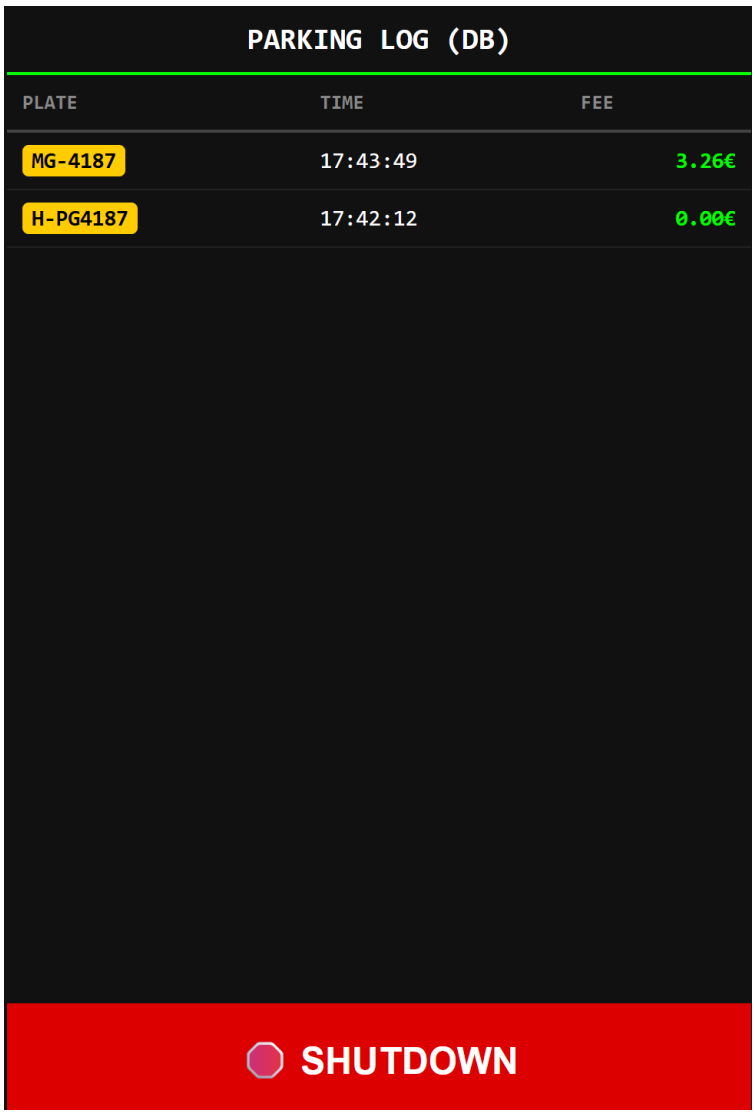


Figure 3.2: The real-time entry/exit log

The billing process automatically determines whether a vehicle should be charged based on its total parking duration. The workflow is divided into four main scenarios as described below.

3.2.1 Scenario 1: Vehicle Entry

- When a car enters the premises, the system automatically detects and records the license plate number.
- The entry time is stored in the database with a timestamp.
- No payment is initiated at this stage.

3.2.2 Scenario 2: Vehicle Exit

- On exit, the system again detects the license plate and retrieves the corresponding entry record.
- The current time is compared with the stored entry timestamp.
- The total parking duration is computed in minutes.

3.2.3 Scenario 3: Duration Less Than 15 Minutes

- If the total parking time is **less than 15 minutes**, the system classifies the session as **free of cost**.
- The display automatically redirects to the **"Danke" (Thank You)** page.
- The barrier opens and the vehicle exits without any payment.

3.2.4 Scenario 4: Duration Greater Than 15 Minutes

- If the parking duration **exceeds 15 minutes**, the system calculates the billing amount at the rate of **€2.00 per hour (pro-rated)**.
- The total payable amount is shown on the display screen.
- The user completes the payment through the integrated terminal.
- Once the transaction is successful, the barrier opens and the system resets for the next vehicle.

Parking Duration	**Billing Amount**
0 – 15 minutes	Free of Cost
> 15 minutes	€2.00 per hour (pro-rated)

Example: If a car remains parked for 42 minutes, the payable amount is calculated as:

$$\text{Amount} = \frac{42}{60} \times 2.00 = 1.40$$

The user is then prompted to make the payment before exit authorization.

3.3 System Power Control

To ensure the longevity of the Jetson Nano hardware and prevent corruption of the local SQLite database, the system must be shut down correctly at the end of the operating hours.

Figure 3.3: The Software Shutdown/Stop Button located on the Dashboard.

Safe Shutdown Procedure:

1. Locate the red **Stop/Shutdown** button at the bottom of the dashboard sidebar.
2. Click the button once; a confirmation dialogue will appear on the screen.
3. Upon confirmation, the system will finalize all database writes and safely power off the Jetson Nano.
4. Only after the screen goes completely dark should the physical power cable be disconnected.

3.4 Operational Environment and Setup

Camera Placement and Setup:

- Mount the camera about **2.0–2.5 metres above ground level**, facing the vehicle lane at a **30–40° angle**.
- Keep a distance of about **3–5 metres** between the camera and the point where vehicles stop or slow down.
- Ensure there is enough ambient light. During night hours, add a small **LED lamp or floodlight** near the gate for better clarity.
- Keep the camera lens clean and free of dust, water spots, or scratches.

Plausibility and Manual Verification:

- If the main unit cannot read a plate clearly (for example, due to dirt, glare, or motion blur), it will display a **"Detection Unclear"** message on the screen.
- In such cases, the operator can manually enter the license plate number using a simple on-screen input field.
- The main unit checks for **duplicate or incomplete entries** to avoid double billing.
- If an entry or exit is missing, the software automatically flags the record for review.

3.5 User Interaction

The system is designed for daily airport use with minimal user effort. All detections, timings, and charge decisions appear automatically on the display.

Interface Overview:

- The screen displays the **camera view** on the left and the **detection log** on the right.
- Each entry shows the plate number, time, duration, and parking status — "Free" (green) or "Charge" (orange).
- Exempt vehicles such as police or ambulance appear in blue with "No Charge."

Daily Operation:

- Power on — the system starts automatically and the AI engine initializes.
- Observe the live feed and alerts for any unreadable plates.
- Use the **Stop Button** (Fig. 3.3) to turn off the display and power after working hours; all data is saved automatically.

This workflow ensures clear, automatic, and reliable operation for non-technical staff, focusing on smooth vehicle flow and accurate billing.

4 Maintenance

4.1 Maintenance

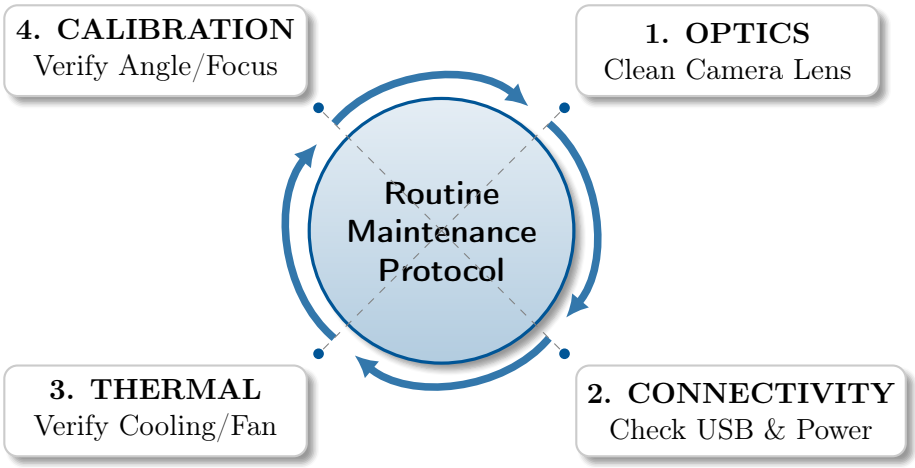
Regular maintenance keeps the **License Plate Detection System** running smoothly and extends its lifetime. This section describes how users can care for the hardware and software to ensure reliable daily operation at the airport gate.

Important

- Turn off power before cleaning or adjusting cables.
- Avoid using sharp objects or sprays near the camera lens.
- Do not block the ventilation openings of the main unit's case.
- If the device overheats or smells unusual, disconnect power immediately and contact support.

4.1.1 Hardware Maintenance

- **Keep it clean:** Gently wipe the main unit and camera with a dry microfiber cloth. Avoid liquids, direct sunlight, or dust build-up around the device.
- **Check connections:** Ensure all USB, HDMI, and power cables are firmly attached. Loose connections can cause flickering displays or camera interruptions.
- **Maintain good cooling:** Place the main unit in a ventilated area. If the device feels unusually hot, turn it off for a few minutes or use a small desk fan.
- **Power stability:** Always use the original 5V/4A USB-C power adapter. Unstable power may lead to sudden restarts or data loss.
- **Camera alignment:** Check every few weeks that the camera faces the correct lane and captures clear plates without obstruction.



4.1.2 Software Maintenance

- **Regular check-ups:** Restart the main unit once every few days to clear memory and refresh processes.
- **Keep the main unit updated:** If an update notification appears, allow the main unit to install recommended main unit updates.
- **Database management:** Every few weeks, check that the detection log is not full. If storage is low, export or delete older records safely.
- **Model refresh (for developers):** Advanced users can retrain or update the detection model to improve recognition accuracy over time.

4.1.3 Backup and Safety

- **Weekly backup:** Copy the database file (`plates.db`) to a USB drive or computer for safekeeping.
- **Power off correctly:** Always shut down the device through the desktop before unplugging power.
- **Avoid sudden shutdowns:** Interrupting the main unit during detection can damage stored data.
- **Emergency reset:** If the screen freezes, disconnect power, wait 10 seconds, and reconnect.
- **Check storage space:** Ensure there is enough free space on the main unit's microSD card. Low storage can slow down performance or prevent new data from being saved.

5 Safety Guidelines

5.1 Safety Guidelines

The following safety guidelines are essential to ensure the reliable, secure, and fault-free operation of the **License Plate Detection System**. Failure to adhere to these protocols may result in hardware damage, data corruption, or safety hazards.

ELECTRICAL

- Use certified 5V/4A Power Supply.
- No hot-plugging (HDMI/Camera).
- Use surge protection devices.
- Inspect cable insulation daily.

THERMAL

- Operational: 0°C to 40°C.
- Do NOT block heatsink vents.
- Active cooling required if > 40°C.
- Keep dry and away from humidity.

OPERATIONAL

- Always use `sudo poweroff`.
- Avoid forced hard reboots.
- Monitor logs during high AI load.
- Disconnect power before transit.

MECHANICAL

- Use ESD Wrist Strap on PCB.
- Secure Camera mounting brackets.
- Prevent cable tension/strain.
- Use stable, level surface.

Figure 5.1: Safety Protocol Matrix: Multi-domain operational constraints.

5.1.1 Electrical Safety

The Jetson Nano is sensitive to voltage fluctuations. Strict adherence to power standards is required:

- **Power Supply:** Always use a certified **5V / 4A USB-C power supply**. Undervoltage will cause the system to shut down unexpectedly during AI inference spikes.
- **Hot-Plugging:** Never connect or disconnect peripherals (Camera, HDMI) while the main unit is live. This can cause transient voltage spikes damaging the board's logic gates.
- **Cabling:** Regularly check cables for fraying. Ensure all connectors are fully seated and not loose.
- **Grounding:** Use surge-protected outlets to prevent static or grid spikes from reaching the board.

5.1.2 Thermal and Mechanical Safety

The system relies on a combination of passive aluminum heatsinks and optional active cooling.

- **Ventilation:** Do not block the vents of the case. Ambient heat buildup can lead to thermal throttling, reducing the FPS of the detection engine.
- **Temperature Range:** Operating range is strictly **0 °C to 40 °C**. If deployed in gatehouses without climate control, active fan cooling is mandatory.
- **Liquids:** The board has no conformal coating; keep all liquids and conductive dust away.
- **ESD Protection:** Static electricity can kill a Maxwell GPU instantly. Use an anti-static wrist strap when opening the case.

5.1.3 Operational Safety

Software-level safety ensures the longevity of the file system and database.

- **Shutdown Procedure:** Sudden power loss can corrupt the SQLite `parking.db` file. Always initiate a shutdown via the dashboard button or terminal.
- **Monitoring:** If the device exhibits abnormal fan noise or LED flickering, power off immediately.
- **Database Integrity:** Ensure database operations are finished before relocating the hardware to avoid write-errors on the SSD.

5.1.4 Inference Safety

Continuous AI processing generates cumulative heat.

- **Load Monitoring:** It is recommended to check the GPU load periodically. Avoid 100% utilization in direct sunlight.
- **Cooling Breaks:** After 24 hours of continuous high-load testing, allow the device a short idle period to stabilize internal temperatures.

6 Troubleshooting

6.1 Troubleshooting

This section lists the most common issues that may occur while operating the **License Plate Detection System**, along with their likely causes and solutions. These checks help maintain main unit stability, reduce downtime, and ensure uninterrupted license plate recognition.

6.1.1 Hardware and Power Issues

- **main unit does not power on:**
  - Confirm the **5V / 4A USB-C adapter** is properly connected and the LED on the main unit is lit.
  - Test a different cable or wall socket.
  - Avoid using unpowered USB hubs to supply power.
- **Device overheats or shuts down unexpectedly:**
  - Ensure proper airflow and use a small fan if necessary.
  - Run `sudo tegrastats` to monitor temperature.
  - Keep ambient temperature below **40 °C**.
- **Camera not detected:**
  - Run `ls /dev/video*` to verify detection.
  - Re-insert the camera or test it on another PC.
  - Use a main USB port, not a low-power one.

6.1.2 Software and Model Issues

- **Low frame rate (FPS):**
  - Use a lighter model (YOLOv8-Nano) or reduce camera resolution to 640×480.
  - Run `sudo nvpmodel -m 0` and `sudo jetson_clocks` for max performance.
  - Close any other GPU-intensive programs.
- **OCR returns wrong text:**

- Clean the camera lens and ensure good lighting.
- Adjust image brightness or threshold parameters.
- Retrain the OCR model with regional plate samples if needed.

6.1.3 Database and Storage Issues

- **Database locked:**
  - Close any parallel process using `plates.db`.
  - Run `sqlite3 plates.db "PRAGMA journal_mode=WAL;"` for safer writes.
  - Restart the app to release file locks.

```
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 0
0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
2023-06-19T21:47:30.166+0300 E STORAGE  [repl writer worker 2] WiredTiger error (-31802) [1687200450:166604][1380581:0x
7f0a4d18e700], file:collection-465--8444901078064748543.wt, WT_CURSOR.search: __wt_block_read_off, 281: collection-465-
-8444901078064748543.wt: fatal read error: WT_ERROR: non-specific WiredTiger error Raw: [1687200450:166604][1380581:0x7
f0a4d18e700], file:collection-465--8444901078064748543.wt, WT_CURSOR.search: __wt_block_read_off, 281: collection-465--
8444901078064748543.wt: fatal read error: WT_ERROR: non-specific WiredTiger error
2023-06-19T21:47:30.166+0300 E STORAGE  [repl writer worker 2] WiredTiger error (-31804) [1687200450:166614][1380581:0x
7f0a4d18e700], file:collection-465--8444901078064748543.wt, WT_CURSOR.search: __wt_panic, 494: the process must exit an
d restart: WT_PANIC: WiredTiger library panic Raw: [1687200450:166614][1380581:0x7f0a4d18e700], file:collection-465--84
44901078064748543.wt, WT_CURSOR.search: __wt_panic, 494: the process must exit and restart: WT_PANIC: WiredTiger librar
y panic
2023-06-19T21:47:30.166+0300 F -        [repl writer worker 2] Fatal Assertion 50853 at src/mongo/db/storage/wiredtiger
/wiredtiger_util.cpp 423
2023-06-19T21:47:30.166+0300 F -        [repl writer worker 2] \n\n***aborting after fassert() failure\n\n
2023-06-19T21:47:30.204+0300 F -        [repl writer worker 2] Got signal: 6 (Aborted).
 0x561c6ea12061 0x561c6ea11279 0x561c6ea1175d 0x7f0a7722cce0 0x7f0a76ea3a4f 0x7f0a76e76db5 0x561c6cf48908 0x561c6d05628
f 0x561c6d0c4009 0x561c6ced0594 0x561c6ced09ae 0x561c6d19a669 0x561c6d19a7b7 0x561c6d0faab1 0x561c6d10183f 0x561c6d125e
genc/ 561c6d0edfda 0x561c6d14395c 0x561c6d034fd5 0x561c6d8bfa5b 0x561c6d880438 0x561c6d896ecb 0x561c6d8bc643 0x561c6d896
ecb 0x561c6d8e89dd 0x561c6d8e936b 0x561c6d83b3b8 0x561c6d7c4af9 0x561c6d7ce5f7 0x561c6d32021b 0x561c6d3239fc 0x561c6d32
46d5 0x561c6d326906 0x561c6d30c074 0x561c6d320785 0x561c6e138d02 0x561c6e13a9b4 0x561c6e13bb36 0x561c6e13c4de 0x561c6eb
22040 0x7f0a772221cf 0x7f0a76e8ed83
----- BEGIN BACKTRACE -----
{"backtrace":[{"b":"561C6C506000","o":"250C061","s":"_ZN5mongo15printStackTraceERSo"},{"b":"561C6C506000","o":"250B279"
},{"b":"561C6C506000","o":"250B75D"},{"b":"7F0A7721A000","o":"12CE0"},{"b":"7F0A76E55000","o":"4EA4F","s":"gsignal"},{"
```

- **Database missing or corrupted:**
  - Restore from backup using `sqlite3 plates.db < backup.sql`.
  - Check free space with `df -h`.
  - Avoid unplugging power during database writes.
- **Storage full:**
  - Delete old logs from `/logs/`.
  - Run `sudo apt clean` to clear unused packages.
  - Expand SD card storage if necessary.

6.1.4 Network or Communication Issues

- **SSH connection fails:**
  - Check IP with `ifconfig` or `hostname -I`.
  - Enable SSH using `sudo system ctl enable ssh`.

  - Inspect router or firewall restrictions.

- **Data not syncing across multiple units:**
  - Confirm all devices have the same network time (`ntpdate`).
  - Test connection via `ping <IP>`.
  - Use a central database or MQTT broker for multi-unit setups.

Troubleshooting Reference Table

Table 6.1: Common Problems, Causes, and Solutions

Problem	**Possible Cause**	**Suggested Solution**
No power or LED off	Faulty adapter or cable	Replace adapter and verify connections
Camera not detected	Loose USB or bad port	Reconnect or test on another PC
Model not loading	Missing weight file	Re-download or verify file path
Low FPS	High resolution / throttling	Reduce resolution / enable performance mode
Database locked	Concurrent access	Restart app / enable WAL mode
OCR errors	Poor lighting or angle	Improve illumination / refocus camera
System overheating	Blocked ventilation	Add small fan or reposition device

Contact for Technical Assistance

For any technical questions or troubleshooting support, please contact the development team:

- Gautam Ramesh – gautam.ramesh.1@stud-hs-emden-leer.com
- Siddhanth Sharma – siddhanth.sharma@stud.hs-emden-leer.de
- Shubhankar Dumka – shubhankar.dumka@stud.hs-emden-leer.de

Response time: typically within **2–3 working days**.

7 Add Ons

7.1 Proposed Add-Ons

The following add-ons can enhance functionality, usability, and scalability of the **License Plate Detection System**. These modules are optional extensions that allow the system to evolve from a standalone edge device into a smart, networked platform for advanced parking, surveillance, or tolling applications.

- **LoRa or Wi-Fi Module for Remote Access:** Enables wireless communication between multiple units or a central dashboard. Modules such as `ESP32` or `Seeed Wio-E5 LoRaWAN` can provide long-range connectivity without relying on wired Ethernet.
- **Edge AI Model Optimization:** Integrate **TensorRT** or **NVIDIA DeepStream SDK** to accelerate YOLOv8 inference and reduce latency under heavy load conditions.
- **Cloud Synchronization:** Enable secure backup and analytics by uploading detections and logs to cloud services such as **AWS S3**, **Firebase**, or **Azure IoT Hub**. This supports remote monitoring and historical trend analysis.
- **Automated Billing and Access Control:** Integrate the database with **RFID readers**, **QR code scanners**, or automatic **barrier gates**. This can automate entry/exit billing in parking or restricted zones.
- **Thermal Monitoring Sensor:** Attach a **DS18B20** or **MLX90614** sensor to track the System's internal temperature. This allows automatic activation of a cooling fan if temperature exceeds safe thresholds.
- **Solar-Powered Operation:** Deploy the entire setup on a **solar panel and battery pack**, making it suitable for outdoor or remote installations where power lines are unavailable.
- **Voice and LED Notification Module:** Add a small **speaker** or **buzzer** with pre-set voice alerts (e.g., "Vehicle detected", "Free parking period exceeded") and LED indicators for system status or errors.
- **AI-Based Anomaly Detection:** Integrate an additional lightweight model to detect abnormal events — such as missing license plates, vehicle tailgating, or unauthorized parking — using pattern recognition.

- **Dashboard Web Interface:** Create a simple **web-based dashboard** (Flask or Streamlit) accessible over LAN/Wi-Fi, showing live camera feed, current detections, and database insights in real time.
- **Multilingual Support:** Add a configuration module that allows operators to switch between **English**, **German**, or other languages in the interface, making it suitable for broader deployment.
- **NFC-Based Admin Login:** Include a small NFC card reader for secure access to configuration and maintenance menus, ensuring only authorized staff can modify system parameters.

These proposed enhancements can transform the system from a fixed gate-level application into a flexible, intelligent, and scalable AI platform — suitable for airports, campuses, or smart-city environments.